

Falsifiable Predictions of a de Sitter–MOND Clock Completion

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Abstract

We collect the sharply falsifiable predictions of a relativistic completion of Milgrom’s dynamics in which the modified-inertia sector is carried by a *clock* — a scalar entering through the clock-projected derivative $Q = n^\mu \partial_\mu \varphi$ — with the acceleration scale locked to the de Sitter/cosmological scale, $a_0 = c^2/2\pi L_{\text{dS}}$, and cosmological dark matter supplied by the conserved Noether charge of the clock-scalar’s shift symmetry. Each prediction is stated as an observable, a threshold, and a discriminator against dark matter or Λ CDM, and is separated into what is robust to the construction and what is kernel-specific. Six are distinctive and currently testable: **(1)** the acceleration scale is *constant in cosmic time* (a flat $a_0(z)$), so the deep-MOND baryonic Tully–Fisher zero-point at $z \simeq 2.5$ is flat (0.00 dex) against a +0.33 dex Λ CDM expectation, and a robustly *rising* $a_0(z)$ (tracking the full $H(z)$, $\sim 3\times$ by $z = 2$) falsifies the theory; **(2)** an *External Field Effect*: internal velocity dispersions of dwarf spheroidals depend on Galactocentric distance, rising $\sim 1.9\times$ across 40–250 kpc, where dark matter predicts *no* dependence — the sharpest discriminator, and the theory reproduces the low dispersion of Crater II; **(3)** in merging clusters the collisionless charge-dust traces the galaxies, so the lensing mass separates from the gas (the Bullet-cluster offset that pure MOND fails); **(4)** the tensor gravitational-wave speed equals c exactly, and a *subdominant, near-luminal scalar polarization* (amplitude set by the clock–metric coupling) is predicted at a level consistent with current LIGO/pulsar-timing bounds; **(5)** the dark matter is a conserved Noether charge and therefore *cannot decay or annihilate*, forbidding indirect detection lines; **(6)** the exact exponential kernel gives a parameter-free $1/r$ correction above the flat rotation velocity, $v^2 = \sqrt{G_b M a_0} + G_b M/4r + 7(G_b M)^{3/2}/96r^2 \sqrt{a_0} + \mathcal{O}(r^{-3})$. Each number is produced by a committed, runnable script; every dimensional quantity is carried on both a_0 footings. **We do not claim the theory is established:** its dark sector carries an intrinsic Big-Bang-Nucleosynthesis fine-tuning (companion paper), and its full relativistic health is a work in progress; these are predictions of a *candidate* class, offered so that data can confirm or kill it. The construction was developed collaboratively in the associated public repository.

1 The class

The predictions below hold for relativistic MOND completions of the schematic form $S = \int \sqrt{-g} [\frac{M^2}{2} R - \Lambda M^2 - K(Q) + F(Q)\Theta + M^2 a_0^2 G(|V|/a_0)] + S_m$, with n_μ the unit gradient of a clock field, $Q = n \cdot \partial \varphi$, $\Theta = \nabla \cdot n$, $G(y) = y^2 + 2(1+y)e^{-y} - 2$ (so $G'(y)/2y = 1 - e^{-y}$), and matter minimally coupled to the single metric g . Two structural facts drive the predictions: (i) a_0 is fixed by Λ , and (ii) the dark matter is the conserved charge $a^3(-K_Q + 3HF_Q) = C$, pressureless and clustering like cold dark matter. Predictions robust to these two facts are marked $\star$; kernel-specific ones are marked $\dagger$.

2 Predictions

P1 (★) — Flat $a_0(z)$; deep-MOND BTFR zero-point at high z . $a_0 \propto \sqrt{\Lambda}$ is constant in cosmic time, so the deep-MOND baryonic Tully–Fisher relation $V^4 = G_b M a_0$ keeps a fixed zero-point with redshift. At $z \simeq 2.5$ the predicted zero-point shift is 0.00 dex, versus a +0.33 dex expectation for a Λ CDM halo-abundance drift. *Falsifier*: a robust a_0 *rising* with z (tracking $H(z)$, $\sim 3\times$ by $z = 2$) kills the Lock. *Test*: a deep-MOND lensed rotator at $z \sim 2\text{--}3$. [L90]

P2 (★) — External Field Effect on dwarf spheroidals. The nonlinear MOND response violates the strong equivalence principle: a dwarf’s internal dispersion is suppressed by the external Galactic field. For a $10^6 M_\odot$ dwarf, σ rises $\sim 1.9\times$ across Galactocentric radius $40 \rightarrow 250$ kpc; Crater II is pulled to $\sigma \simeq 2.4 \text{ km s}^{-1}$ (observed $\simeq 2.7$). *Discriminator*: dark matter predicts *no* $\sigma\text{--}R_{\text{gc}}$ dependence. This is the sharpest MOND-vs-dark-matter test. [L89]

P3 (★) — Bullet-type mergers. The dark matter is a collisionless conserved charge, so in a cluster merger the lensing mass separates from the X-ray gas and traces the galaxies — the offset pure MOND famously fails. A centroid model with the dust at $\sim 6\times$ the baryons places the lensing peak on the galaxy side ($x_c/d \simeq 0.15\text{--}0.36$ vs 0.93 for pure MOND-on-gas). *Discriminator*: pure MOND fails; this class passes qualitatively. [L85]

P4 (★) — Gravitational-wave speed and polarization. $c_T = c$ exactly (the scalar/MOND terms contribute no graviton kinetic term), passing GW170817 structurally. A subdominant, near-luminal scalar polarization (breathing/longitudinal, mass $\sim 10^{-33}$ eV, amplitude $\propto F_Q$) is predicted, consistent with current LIGO/Virgo/pulsar-timing polarization bounds. *Falsifier*: any measured $c_T \neq c$; *discriminator*: detection of the scalar mode. [L88]

P5 (★) — No dark-matter decay or annihilation. The dark matter is the Noether charge of an exact shift symmetry, hence exactly conserved. The class predicts *no* decay lines and *no* annihilation signal, unlike a WIMP/particle relic. *Falsifier*: a confirmed dark-matter decay or annihilation detection. [L81]

P6 (†) — A $1/r$ correction above the flat rotation velocity. Inverting the exact implicit law gives, for a point mass,

$$v^2 = \sqrt{G_b M a_0} + \frac{G_b M}{4r} + \frac{7(G_b M)^{3/2}}{96 r^2 \sqrt{a_0}} + \mathcal{O}(r^{-3}),$$

a parameter-free correction with no fitting freedom. *Discriminator*: a dark-matter halo gives a profile-dependent (fittable) correction; this one is fixed by the kernel. *Test*: precise rotation curves in the transition region $g \sim a_0$. [L94]

3 Consistency checks (must-match, not distinctive)

The class must also reproduce, and does at the levels tested: the SPARC radial-acceleration relation (exponential kernel, rms 0.16 dex at zero per-galaxy parameters, ranking just below the ν_{RAR} kernel; [L92]); the linear structure-growth amplitude ($\sigma_8 = 0.811$, matching Λ CDM to machine precision because the charge-dust has $c_s^2 = 0$ and standard G_{eff} ; [L93]); galaxy weak-lensing following the same relation as dynamics (no-slip $\Phi = \Psi$); and the Solar-System post-Newtonian bounds, including the preferred-frame parameters that exclude the earlier AeST completion (α_1 is structurally suppressed here rather than un-tunable; [L91]).

4 Honest scope

These are predictions of a candidate class, not of an established theory. Two costs are stated in the companion work: the charge-dust carries an intrinsic ~ 24 -order Big-Bang-Nucleosynthesis fine-tuning (the same charge sources an a^{-6} stiff partner), and the full relativistic constraint algebra / health of the propagating sector is still being derived. The wide-binary signal predicted by the exponential kernel ($\gamma_v \simeq 1.03\text{--}1.06$) sits below the strongest claimed detections and is consistent with the Newton-leaning analyses, so wide binaries chiefly test *which* interpolating kernel is correct rather than MOND versus dark matter. No observation is used here to assert the theory is true; the measured quantities run only the other way, saying how large an effect a member of the class must show and what would falsify it. Every number is produced by a committed script with checks that can fail (`fable_independent_2026/L81,L85,L88-L94`); dimensional numbers are carried on both a_0 footings ($9.3619 \times 10^{-11} \text{ m s}^{-2}$, $1.1279 \times 10^{-10} \text{ m s}^{-2}$). AI-assisted research draft; not peer reviewed.

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